Chronos: Sequential Identity-Preserving Video Generation by Temporal Anchoring

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Introduction

Background

- Reference images specify **who should appear**. Action descriptions specify **what should happen and, in a sequence, when**.
- The same person can look very different across poses, viewpoints, and occlusions. Identity must remain recognizable as the action unfolds

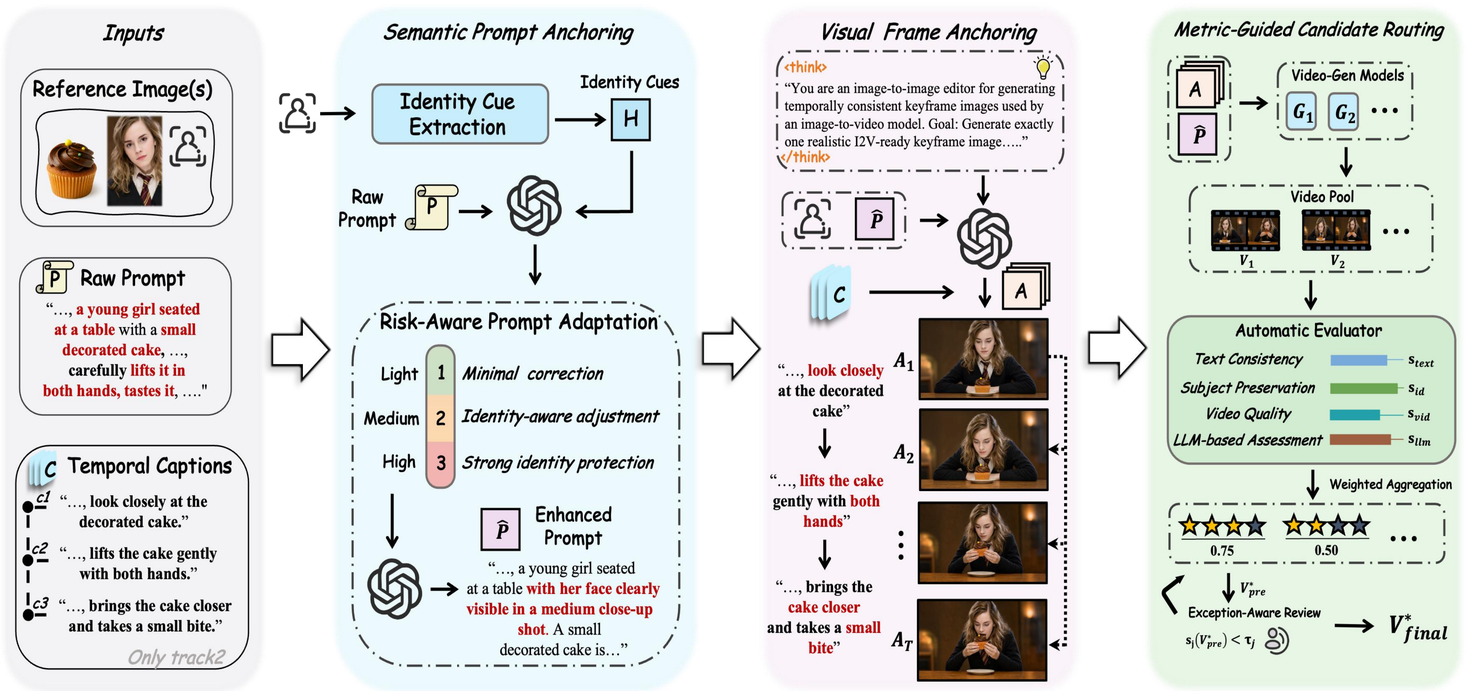
Motivation

- ***Existing limitation.*** Strong identity constraints can suppress motion, while weak constraints allow identity drift.
- ***Our insight.*** Identity-consistent keyframes show how the subject should appear at key action stages, leaving the model free to generate the transitions.

Contribution

- **Concept.** ***Chronos*** introduces a training-free temporal anchoring framework that unifies single-segment facial identity preservation and multi-segment sequential action generation.
- **Technology.** ***Semantic Prompt Anchoring*** adjusts viewpoint and visibility cues according to identity-drift risk. ***Visual Frame Anchoring*** generates a first-frame anchor for a single segment or builds successive keyframes from earlier ones for an action sequence. ***Metric-Guided Candidate Routing*** selects among generators' outputs based on identity, action alignment, and visual quality.
- **Experiments.** In the *ACM MM 2026* Identity-Preserving Video Generation Challenge, ***Chronos*** tied for first in Track 1 and placed first outright in Track 2. Its total scores exceeded the strongest baseline on each track by **9.0%** and **6.6%**, respectively.

Proposed Method



Semantic Prompt Anchoring (SPA)

- SPA extracts **stable identity cues** from reference images, omitting transient details such as clothing and background.
- When distance, viewpoint, or occlusion makes the subject hard to recognize, SPA refines **viewpoint and visibility cues** in the prompt while preserving the intended scene and action.

Visual Frame Anchoring (VFA)

- For a **single segment**, VFA combines the reference images and refined prompt to generate a first frame that places the subject in the requested scene.
- For an **action sequence**, VFA uses each segment's action caption and earlier keyframes to generate the next anchor, preserving identity as the action changes.

Metric-Guided Candidate Routing (MGCR)

- MGCR generates candidate videos with multiple models and scores each for **identity preservation, text alignment, motion, and visual quality**.
- It selects the candidate with the **highest weighted score** and flags it for human review if any critical metric falls below its threshold.

Experiments and Analysis

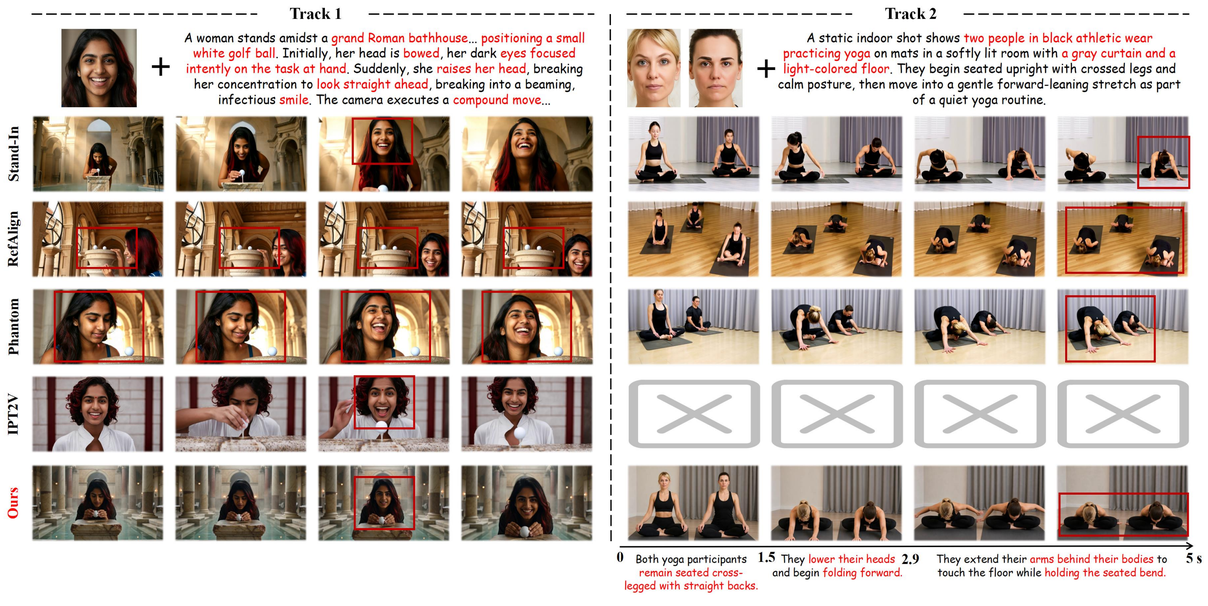
Comparison with existing methods

Method	Identity Consistency	Text Alignment		Video Quality				Total Score↑
	Subject Score↑	GMEScore↑	GMEScore_Seg↑	Aesthetic Score↑	Motion Amplitude↑	Motion Smoothness↑	Natural Score↑	
Track 1								
Stand-In-14B [44]	0.2396	0.6286	-	0.4878	**0.4589**	0.9112	0.5550	0.5042
RefAlign-14B [39]	**0.7996**	0.6342	-	0.4758	0.2946	0.9637	0.4837	0.6214
Phantom-14B [27]	0.7165	0.6242	-	0.5041	0.2102	0.8882	0.5712	0.6211
IPT2V-14B [41]	0.5387	0.5572	-	0.4940	0.3864	0.9004	0.5387	0.5614
USTC-CMI(Chronos)	0.6840	**0.6411**	-	**0.5984**	0.4306	**0.9709**	**0.6725**	**0.6769**
Track 2								
Stand-In-14B [44]	0.1844	**0.7268**	**0.5663**	0.4836	**0.3027**	0.9191	0.6350	0.5100
RefAlign-14B [39]	0.3877	0.6996	0.5392	0.4930	0.1620	0.9838	0.5812	0.5430
Phantom-14B [27]	0.4136	0.7170	0.5432	0.5268	0.2110	0.9613	0.6387	0.5725
USTC-CMI(Chronos)	**0.4400**	0.6907	0.5407	**0.6009**	0.1479	**0.9950**	**0.70625**	**0.6102**

Ablation Study of Modules

Method	Identity Consistency	Text Alignment		Video Quality				Total Score↑
	Subject Score↑	GMEScore↑	GMEScore-Seg↑	Aesthetic Score↑	Motion Amplitude↑	Motion Smoothness↑	Natural Score↑	
Track 1								
baseline	0.7511	0.6188	-	0.5001	0.1827	0.9548	0.5775	0.6353
+ SPA	**0.7557**	0.6176	-	0.5030	0.1763	0.9529	0.5875	0.6394
+ SPA & VFA (use Seedance)	0.6736	0.6357	-	0.5824	**0.4996**	0.9645	0.6475	0.6646
+ SPA & VFA (use Wan)	0.6816	**0.6650**	-	0.6000	0.3652	0.9508	0.5800	0.6487
USTC-CMI(Chronos)	0.6981	0.6475	-	**0.6063**	0.4798	**0.9762**	**0.6675**	**0.6833**
Track 2								
baseline	0.4576	**0.6872**	**0.5504**	0.5278	0.1893	0.9632	0.6300	0.5803
+ SPA & VFA (use Wan I2V-14B)	0.5338	0.6741	0.5502	0.6131	**0.1963**	0.9825	0.6450	0.6177
+ SPA & VFA (use Wan FLF2V-14B)	0.5911	0.6771	0.5473	**0.6439**	0.1505	**0.9956**	0.6625	0.6417
USTC-CMI(Chronos)	**0.6085**	0.6761	0.5496	0.6406	0.1743	0.9944	**0.6850**	**0.6530**

Comparison with existing methods



Challenge Results

Track 1: Facial Identity-Preserving			Track 2: Sequential Action Identity-Preserving		
Team	Final Score↓	Rank	Team	Final Score↓	Rank
USTC-CMI	2.5	1	**USTC-CMI**	1.25	1
MIPL_Video	2.5	1	WislabGDUT	1.8125	2
xuanyuan_fuxi	3	3	USTC-AC	2.9375	3

Key Findings

- **Track 1.** Chronos achieves the highest text alignment and scores **~9% higher overall** than RefAlign.
- **Track 2.** Chronos achieves the highest subject consistency and scores **6.6% higher overall** than Phantom.
- **Ablation.** Metric-guided candidate routing improves the total score over the best single generator on **both tracks**.